

PENGJU LIU

✉ liu03486@umn.edu 🌐 Homepage

RESEARCH INTERESTS

- **AI Agents for Electronic Design Automation (EDA)**
- **Self-Improving Agents**
- **World Models and Predictive Representation Learning (e.g., JEPA)**

EDUCATION

University of Minnesota Twin Cities 2025 – Present

Ph.D. in Computer Science Advisor: Prof. Caiwen Ding

Hong Kong University of Science and Technology 2023 – 2024

Master's degree, Department of Electronic and Computer Engineering Advisor: Prof. Wei Zhang

University of Electronic Science and Technology of China 2019 – 2023

Bachelor's degree in Microelectronics

WORK EXPERIENCE

Shanghai Jiao Tong University 2024 – 2025

Research Assistant Supervisor: Prof. Zhezhi He

- Focused on using large language models (LLMs) for Verilog code generation.

FIRST-AUTHOR PUBLICATIONS

*Equal contribution.

Bridging the Last Mile of Circuit Design: PostEDA-Bench, a Hierarchical Benchmark for PPA Convergence and DRC Fixing

Pengju Liu, Nuo Xu, Jinwei Tang, Yu Cao, and Caiwen Ding. *Under review*, 2026. [Paper] [Code]

- Introduced a hierarchical benchmark of 145 tasks for evaluating LLM agents on design rule checking (DRC) repair and power, performance, and area (PPA) optimization.
- Evaluated eight LLMs across agent frameworks, identifying geometric reasoning and PPA trade-offs as key challenges; adding vision improved DRC repair.

AgentDRC: Multi-Agent Algorithms for Spatially-Aware Automated DRC Correction

Pengju Liu, Nuo Xu, Winson Chen, Peijun Li, Yang (Katie) Zhao, Yu Cao, Caiwen Ding *Under review*, 2026.

HaVen: Hallucination-Mitigated LLM for Verilog Code Generation Aligned with HDL Engineers

Yiyao Yang*, Fu Teng*, **Pengju Liu***, Mengnan Qi, Chenyang Lv, Ji Li, Xuhong Zhang, and Zhezhi He.

Design, Automation & Test in Europe (DATE), 2025. [Paper] [Code]

- Developed a framework to reduce symbolic, knowledge, and logical hallucinations in Verilog generation through chain-of-thought reasoning and targeted data augmentation.
- Aligned generated code with HDL engineering practices, improving functional correctness on the VerilogEval and RTLLM benchmarks.

HONORS AND AWARDS

DAC Young Fellowship 2026

TEACHING ASSISTANT

CSCI 4061: Introduction to Operating Systems Fall 2026

University of Minnesota Twin Cities